UX RESEARCH · E-WALLET

Understanding everyday mobile payers

Goal: learn how people in Thailand pay, transfer and top up by phone, what makes them trust a wallet, and where current apps let them down — then turn that into the design of S1–S26.

Status: research plan + early hypotheses. Findings below are assumptions from desk research, to be validated with the interviews and tests listed.

1 Research questions

- How do people choose between a bank app and an e-wallet for daily payments?
- What makes a payment feel safe — and what makes people abort?
- Where do new users drop off during sign-up and identity checks?
- What do people need to see after paying (slip, status, balance)?
- How do users want to track spending?

2 Methods

Method	Scope	Focus
1:1 interviews	8 participants · 45 min	Daily payment habits, trust, pain points with current apps
Online survey	Target 100+ responses	How often people pay by QR, transfer and top up; what stops them
Competitive review	5 wallet & banking apps	Onboarding steps, home layout, confirm & error patterns
Usability test	5 users · prototype S1–S26	Task success: sign up, transfer, scan, pay a bill

3 Key insights → design response

Speed beats features · Hypothesis
People open a wallet to do one thing — pay, transfer or check balance — and leave within seconds.
Design response: Home: balance + 4 quick actions above the fold · **S7**

Trust is earned at confirm · Hypothesis
The biggest fear is sending to the wrong person or the wrong amount.
Design response: Name + photo on confirm, Face ID, e-slip · **S11 · S12**

Sign-up drop-off · Hypothesis
Identity checks that fail in poor light and restart from zero make people quit.
Design response: 4-step progress bar, live photo guide, saved steps · **S2–S5**

Errors need a way out · Hypothesis
A “failed” message with no next step feels like lost money.
Design response: Every error offers a fix: top up, use PIN, scan again · **S17–S19**

Hidden fees kill trust · Hypothesis
People hesitate when fees and arrival time are not shown before paying.
Design response: Fee and arrival time shown before confirm · **S11 · S23**

Money awareness · Hypothesis
Users want to know where the month’s money went without doing maths.
Design response: Monthly spending, budget bar, in/out filters · **S7 · S15**

4 Competitive review

	Onboarding	Home layout	Confirm step	Error recovery	Spending insight
Bank apps	Long, many forms	Busy, many menus	PIN + OTP	Generic error	Basic list
Super-app wallets	Fast	Crowded with promos	PIN	Some guidance	Limited
Shopping wallets	Fast, inside shop	Promo-first	Auto-pay	Weak	Order-based
Our wallet (target)	4 clear steps	Balance + 4 actions	Face ID + recipient photo	Fix on every error	Budget + monthly view

5 Success metrics

Sign-up completion	% who finish S1 → S5
Time to first payment	From install to first transfer or scan
Transfer success rate	Confirmed without error or retry
Error recovery rate	Users who finish after S17–S19
SUS score	Usability test, target ≥ 80

6 Design principles

Fast	Any key task in 3 taps or fewer
Clear	One main action per screen, plain words
Safe	Show who, how much and fees before paying
Forgiving	Every error has a next step

USER PERSONAS · E-WALLET

Who we are designing for

Three proto-personas built from the research hypotheses. Mint is the primary persona used in the Journey Map; refine all three after interviews.

M · Mint Srisuk

Primary persona

27 · Office worker · Bangkok

> “I just want to pay and go — but I need to know it went to the right person.”

GOALS
- Pay for coffee, food and transport in seconds
- Send money to family every month
- Keep an eye on monthly spending

FRUSTRATIONS
- Bank apps are slow and full of menus
- Afraid of sending to the wrong account
- Sign-up that fails and restarts

BEHAVIOURS
- Pays by QR 5–10 times a day
- Uses Face ID for everything
- Checks balance before big payments

Tech comfort
Payment frequency
Security concern

Key screens: S1–S12 · S15

S · Somchai Wongdee

Secondary persona

52 · Food stall owner · Chiang Mai

> “I need to see the money arrive, clearly, before the customer walks away.”

GOALS
- Receive payments with his own QR
- See every incoming payment instantly
- Withdraw to his bank at the end of the day

FRUSTRATIONS
- Small text and too many icons
- Unsure whether a payment really arrived
- Scared of tapping the wrong button

BEHAVIOURS
- Shows printed QR at the stall
- Checks notifications after each sale
- Withdraws daily to his bank

Tech comfort
Payment frequency
Security concern

Key screens: S14 · S23 · S25

P · Ploy Kittikun

Secondary persona

19 · University student · Khon Kaen

> “My allowance has to last the month — I want to see where it goes.”

GOALS
- Top up phone credit and data quickly
- Split bills with friends
- Stay within a monthly budget

FRUSTRATIONS
- Runs out of balance at the counter
- Can’t tell what she spent on
- Hidden fees on top-ups

BEHAVIOURS
- Gets allowance by transfer
- Tops up data every week
- Pays friends back the same day

Tech comfort
Payment frequency
Security concern

Key screens: S8 · S17 · S22

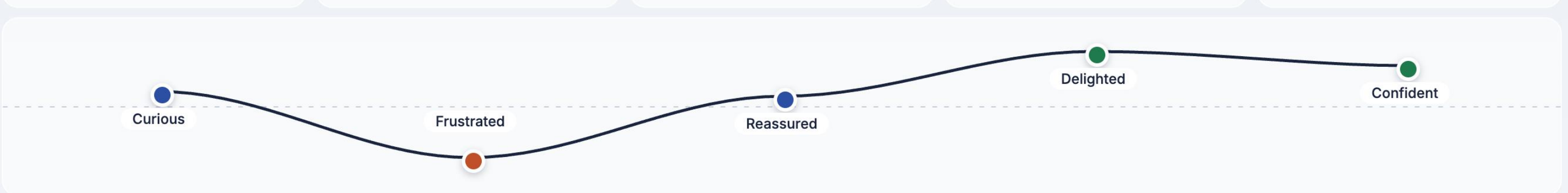
CUSTOMER JOURNEY MAP

Everyday payments & transfers with an e-wallet

Scenario: a new user wants to pay for coffee and send money to family quickly — without thinking too hard, and feeling sure it's safe.

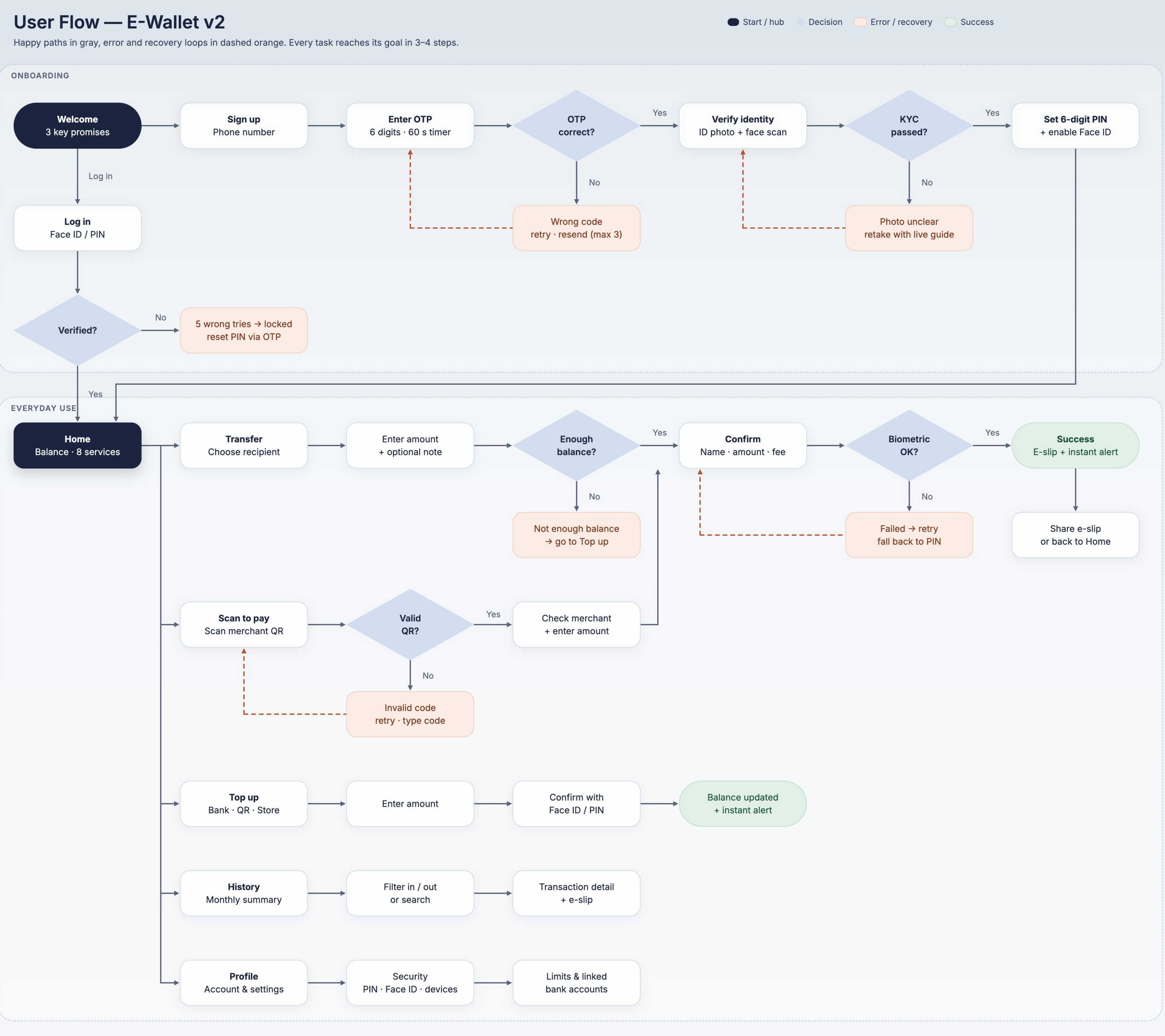
Persona
M · Mint, 27 · Office worker, Bangkok
Pays by phone every day. Wants speed, clarity and security; dislikes complicated steps.

	1 Discover	2 Sign up & verify	3 Top up	4 Pay / Transfer	5 Track & return
Goals	Find a faster, easier way to pay than cash	Open an account in a few minutes	Have money ready to spend right away	Pay or transfer in 3 taps	Know where this month’s money went
Actions	Sees an ad / friend recommends it Reads reviews, downloads the app	Enters phone · confirms OTP Photographs ID · face scan Sets a 6-digit PIN	Links a bank account or tops up via QR	Scans merchant QR / picks a recipient Enters amount · confirms with Face ID	Opens history · filters in / out Checks the monthly summary
Touchpoints	Social media · App Store	Sign-up screens · camera · SMS	Home · notifications	Scan · Transfer · confirm screen	History · summary notifications
Thoughts	“Is it safe? Are there fees?”	“So many steps... the ID photo failed again.”	“Has the money arrived yet?”	“That was fast! But is the name right?”	“Did I spend too much this month?”
Emotion positive ↑ / negative ↓	Curious	Frustrated	Reassured	Delighted	Confident
Pain points	Fees and security info are unclear	Identity check fails in low light and restarts from scratch	No clear status on when money arrives	Fear of sending to the wrong person; confirm button sits too close to others	Long history, hard to find a specific transaction
Opportunities	Welcome screen with 3 key promises: no fees · secure · fast	Progress bar, auto-saved steps and live photo guidance	Balance front and center on Home + instant top-up alert	Clear recipient name and photo, quick amount chips, biometric confirm	Monthly in / out summary + 3-way filter



User Flow — E-Wallet v2

Happy paths in gray, error and recovery loops in dashed orange. Every task reaches its goal in 3–4 steps.



Start / hub · Decision · Error / recovery · Success

ONBOARDING

Welcome 3 key promises → **Sign up** Phone number → **Enter OTP** 6 digits · 60 s timer → **OTP correct?** Yes → **Verify identity** ID photo + face scan → **KYC passed?** Yes → **Set 6-digit PIN** + enable Face ID

OTP correct? No → Wrong code retry · resend (max 3)

KYC passed? No → Photo unclear retake with live guide

Welcome → Log in → **Log in** Face ID / PIN → **Verified?** No → 5 wrong tries → locked reset PIN via OTP

Verified? Yes → Home

EVERYDAY USE

Home Balance · 8 services → **Transfer** Choose recipient → Enter amount + optional note → **Enough balance?** Yes → **Confirm** Name · amount · fee → **Biometric OK?** Yes → **Success** E-slip + instant alert → Share e-slip or back to Home

Enough balance? No → Not enough balance → go to Top up

Biometric OK? No → Failed → retry fall back to PIN

Scan to pay Scan merchant QR → **Valid QR?** Yes → Check merchant + enter amount → Confirm

Valid QR? No → Invalid code retry · type code

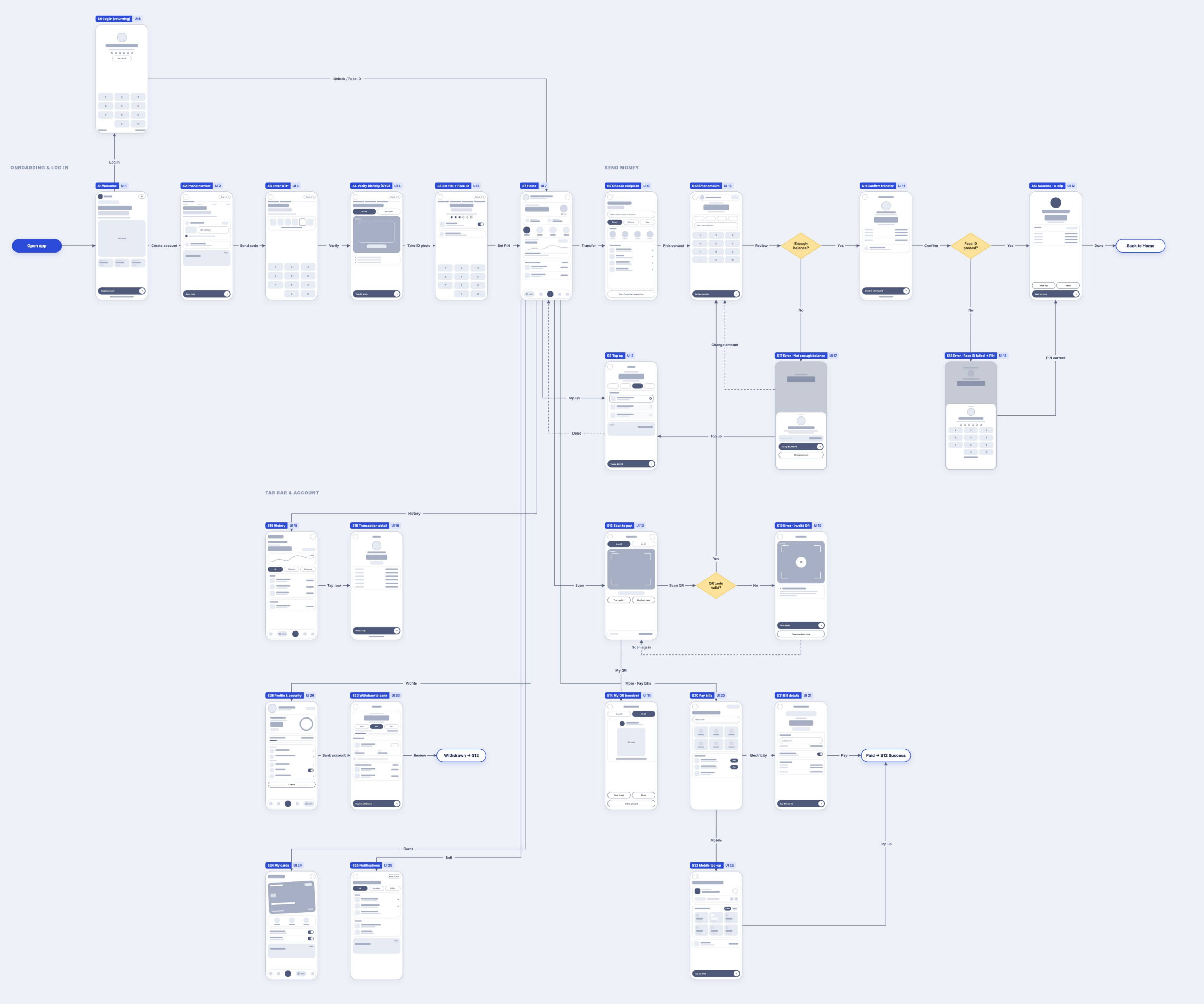
Top up Bank · QR · Store → Enter amount → Confirm with Face ID / PIN → Balance updated + instant alert

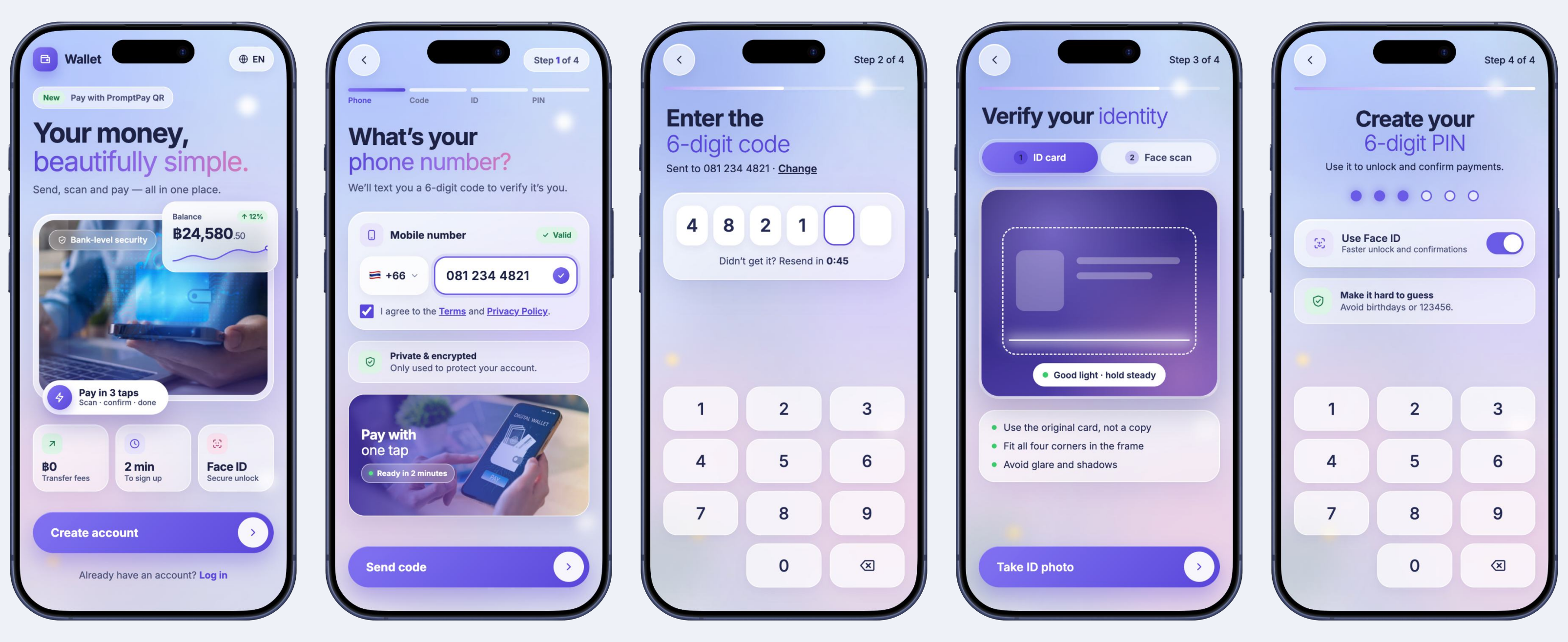
History Monthly summary → Filter in / out or search → Transaction detail + e-slip

Profile Account & settings → Security PIN · Face ID · devices → Limits & linked bank accounts

Wireframe User Flow

E-Wallet · 26 low-fidelity wireframes connected by tap paths: onboarding, log in, send money, scan to pay, top up, bills, withdraw and account. Badges show the wireframe number and its UI frame.





Wallet · EN
New · Pay with PromptPay QR
Your money, beautifully simple.
Send, scan and pay — all in one place.
Bank-level security · Balance ฿24,580.50 · ↑ 12%
Pay in 3 taps · Scan · confirm · done
฿0 Transfer fees · 2 min To sign up · Face ID Secure unlock
Create account
Already have an account? Log in

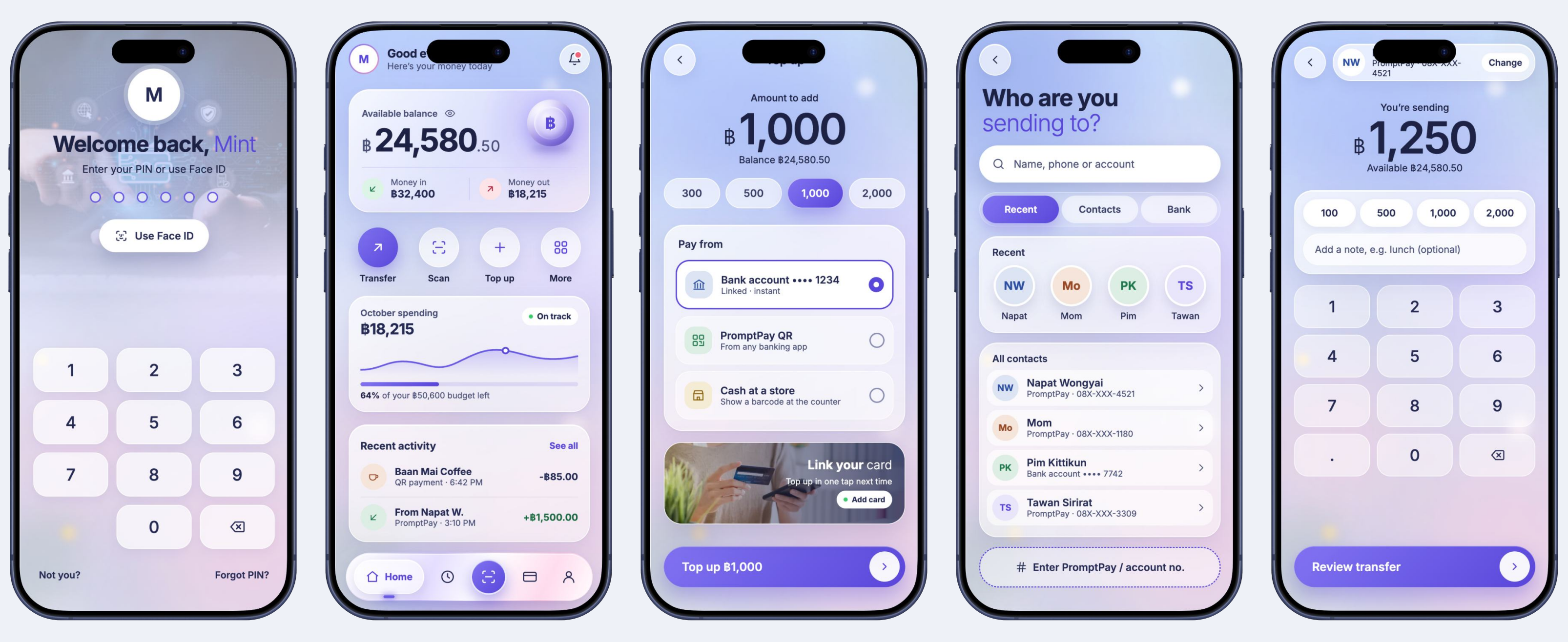
Step 1 of 4 · Phone · Code · ID · PIN
What’s your phone number?
We’ll text you a 6-digit code to verify it’s you.
Mobile number ✓ Valid · +66 · 081 234 4821
☑ I agree to the Terms and Privacy Policy.
Private & encrypted · Only used to protect your account.
Pay with one tap · Ready in 2 minutes
Send code

Step 2 of 4
Enter the 6-digit code
Sent to 081 234 4821 · Change
4 8 2 1
Didn’t get it? Resend in 0:45

Step 3 of 4
Verify your identity
1 ID card · 2 Face scan
Good light · hold steady
- Use the original card, not a copy
- Fit all four corners in the frame
- Avoid glare and shadows

Take ID photo

Step 4 of 4
Create your 6-digit PIN
Use it to unlock and confirm payments.
Use Face ID · Faster unlock and confirmations
Make it hard to guess · Avoid birthdays or 123456.



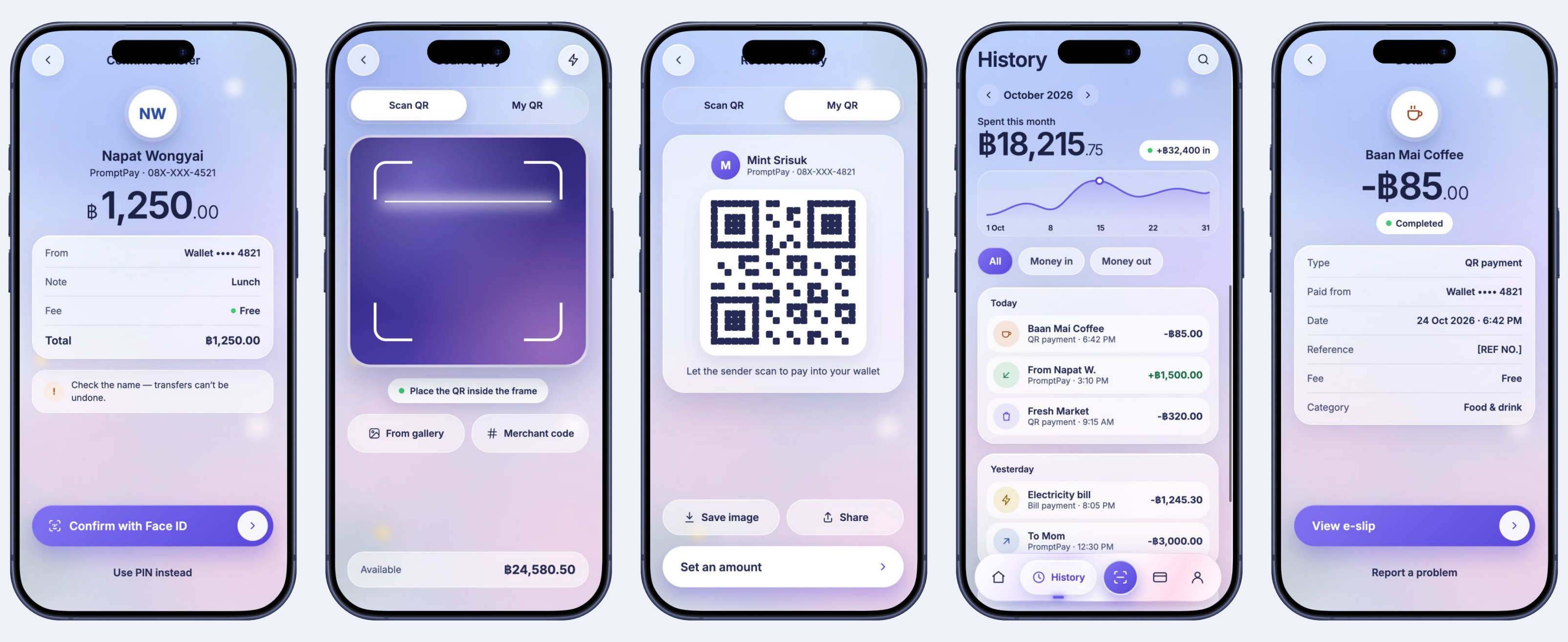
Welcome back, Mint
Enter your PIN or use Face ID
Use Face ID
Not you? · Forgot PIN?

Here’s your money today
Available balance ฿24,580.50
Money in ฿32,400 · Money out ฿18,215
Transfer · Scan · Top up · More
October spending ฿18,215 · On track
64% of your ฿50,600 budget left
Recent activity · See all
Baan Mai Coffee · QR payment · 6:42 PM · -฿85.00
From Napat W. · PromptPay · 3:10 PM · +฿1,500.00
Home

Amount to add ฿1,000
Balance ฿24,580.50
300 · 500 · 1,000 · 2,000
Pay from
Bank account •••• 1234 · Linked · instant
PromptPay QR · From any banking app
Cash at a store · Show a barcode at the counter
Link your card · Top up in one tap next time · Add card
Top up ฿1,000

Who are you sending to?
Name, phone or account
Recent · Contacts · Bank
Recent: Napat · Mom · Pim · Tawan
All contacts
Napat Wongyai · PromptPay · 08X-XXX-4521
Mom · PromptPay · 08X-XXX-1180
Pim Kittikun · Bank account •••• 7742
Tawan Sirirat · PromptPay · 08X-XXX-3309
Enter PromptPay / account no.

Change
You’re sending ฿1,250
Available ฿24,580.50
100 · 500 · 1,000 · 2,000
Add a note, e.g. lunch (optional)
Review transfer



Napat Wongyai · PromptPay · 08X-XXX-4521
฿1,250.00

From	Wallet •••• 4821
Note	Lunch
Fee	Free
Total	฿1,250.00

Check the name — transfers can’t be undone.
Confirm with Face ID
Use PIN instead

Scan QR · My QR
Place the QR inside the frame
From gallery · Merchant code
Available ฿24,580.50

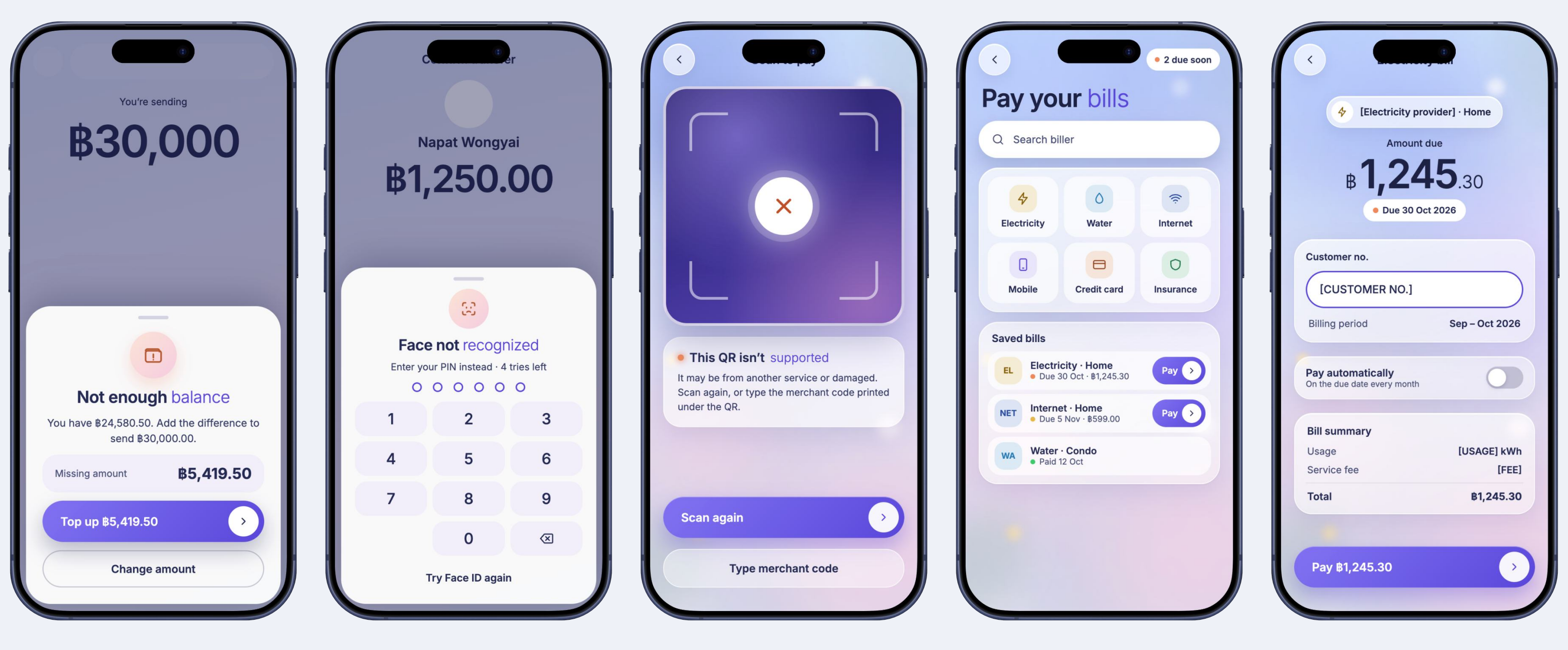
Scan QR · My QR
Mint Srisuk · PromptPay · 08X-XXX-4821
Let the sender scan to pay into your wallet
Save image · Share
Set an amount

History
October 2026
Spent this month ฿18,215.75 · +฿32,400 in
All · Money in · Money out
Today
Baan Mai Coffee · QR payment · 6:42 PM · -฿85.00
From Napat W. · PromptPay · 3:10 PM · +฿1,500.00
Fresh Market · QR payment · 9:15 AM · -฿320.00
Yesterday
Electricity bill · Bill payment · 8:05 PM · -฿1,245.30
To Mom · PromptPay · 12:30 PM · -฿3,000.00

Baan Mai Coffee
-฿85.00
Completed

Type	QR payment
Paid from	Wallet •••• 4821
Date	24 Oct 2026 · 6:42 PM
Reference	[REF NO.]
Fee	Free
Category	Food & drink

View e-slip
Report a problem



You’re sending ฿30,000
Not enough balance
You have ฿24,580.50. Add the difference to send ฿30,000.00.
Missing amount ฿5,419.50
Top up ฿5,419.50
Change amount

Napat Wongyai ฿1,250.00
Face not recognized
Enter your PIN instead · 4 tries left
Try Face ID again

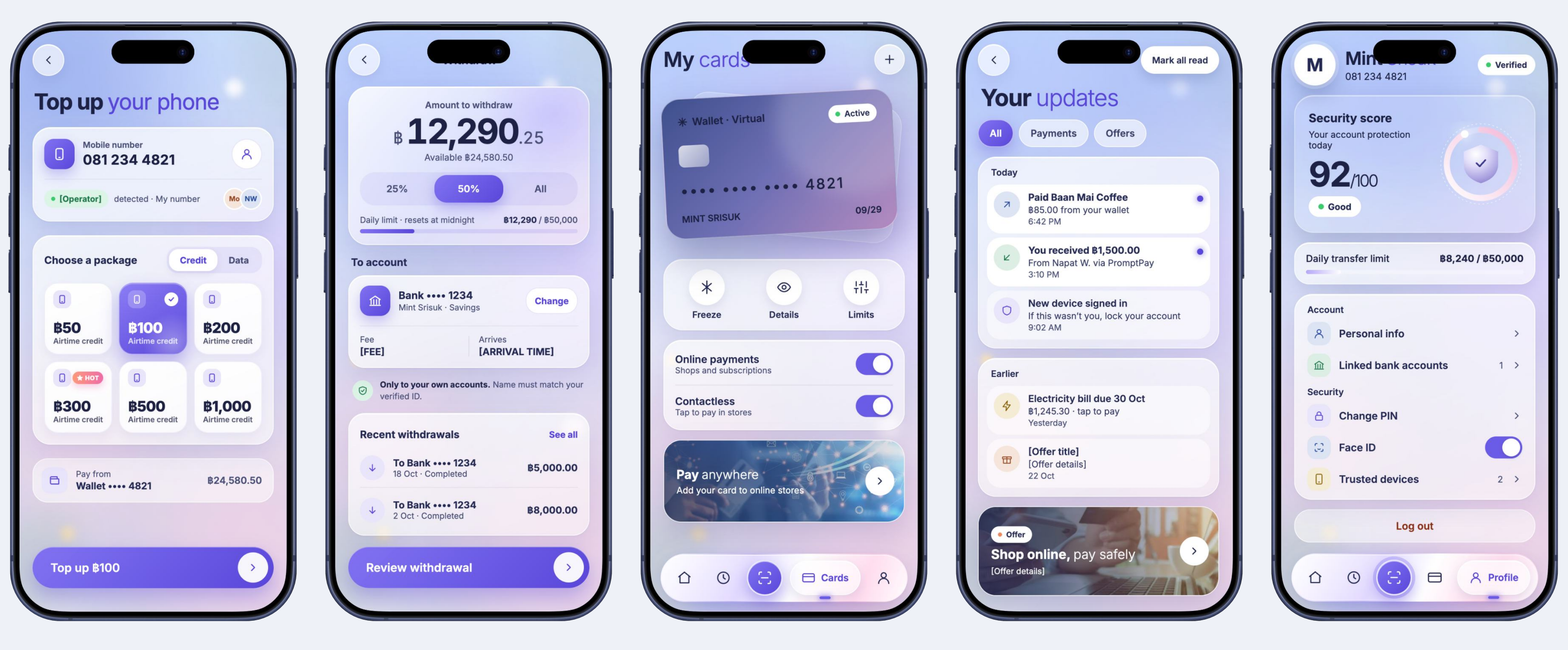
This QR isn’t supported
It may be from another service or damaged. Scan again, or type the merchant code printed under the QR.
Scan again
Type merchant code

2 due soon
Pay your bills
Search biller
Electricity · Water · Internet · Mobile · Credit card · Insurance
Saved bills
Electricity · Home · Due 30 Oct · ฿1,245.30 · Pay
Internet · Home · Due 5 Nov · ฿599.00 · Pay
Water · Condo · Paid 12 Oct

[Electricity provider] · Home
Amount due ฿1,245.30
Due 30 Oct 2026
Customer no. [CUSTOMER NO.]
Billing period Sep – Oct 2026
Pay automatically · On the due date every month
Bill summary

Usage	[USAGE] kWh
Service fee	[FEE]
Total	฿1,245.30

Pay ฿1,245.30



Top up your phone
Mobile number 081 234 4821
[Operator] detected · My number
Choose a package · Credit · Data
฿50 Airtime credit · ฿100 Airtime credit · ฿200 Airtime credit
HOT ฿300 Airtime credit · ฿500 Airtime credit · ฿1,000 Airtime credit
Pay from Wallet •••• 4821 · ฿24,580.50
Top up ฿100

Amount to withdraw ฿12,290.25
Available ฿24,580.50
25% · 50% · All
Daily limit · resets at midnight · ฿12,290 / ฿50,000
To account
Bank •••• 1234 · Mint Srisuk · Savings · Change
Fee [FEE] · Arrives [ARRIVAL TIME]
Only to your own accounts. Name must match your verified ID.
Recent withdrawals · See all
To Bank •••• 1234 · 18 Oct · Completed · ฿5,000.00
To Bank •••• 1234 · 2 Oct · Completed · ฿8,000.00
Review withdrawal

My cards
Wallet · Virtual · Active · •••• •••• •••• 4821 · MINT SRISUK · 09/29
Freeze · Details · Limits
Online payments · Shops and subscriptions
Contactless · Tap to pay in stores
Pay anywhere · Add your card to online stores
Cards

Mark all read
Your updates
All · Payments · Offers
Today
Paid Baan Mai Coffee · ฿85.00 from your wallet · 6:42 PM
You received ฿1,500.00 · From Napat W. via PromptPay · 3:10 PM
New device signed in · If this wasn’t you, lock your account · 9:02 AM
Earlier
Electricity bill due 30 Oct · ฿1,245.30 · tap to pay · Yesterday
[Offer title] · [Offer details] · 22 Oct
Offer · Shop online, pay safely · [Offer details]

081 234 4821 · Verified
Security score · Your account protection today · 92/100 · Good
Daily transfer limit ฿8,240 / ฿50,000
Account
Personal info
Linked bank accounts 1
Security
Change PIN
Face ID
Trusted devices 2
Log out
Profile